

COMPSCI 689

Lecture 19: Latent Factor Models

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Multivariate Normal

- The multivariate normal (or Gaussian) distribution is a fundamental building block for unsupervised learning with vector-valued random variables $\mathbf{X} \in \mathbb{R}^D$.
- The distribution has two parameters $\theta = [\mu, \Sigma]$. μ is the mean vector and Σ is the covariance matrix.
- The probability density is given below (assuming $\mathbf{x}$ and μ are column vectors):

$$\mathcal{N}(\mathbf{x}; \mu, \Sigma) = \frac{1}{|2\pi\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu)\right)$$

- We have $\mu \in \mathbb{R}^D$ and $\Sigma \in \mathbb{S}_+^D$, the space of symmetric, positive definite $D \times D$ matrices.

Special Cases of MVNs

- If $\mu = [0, \dots, 0]$ and $\Sigma = I$ is the identity matrix, $\mathcal{N}(\mathbf{x}; \mu, \Sigma)$ is called a “standard multivariate normal” or “standard Gaussian.”
- If $\Sigma = \sigma I$ for $\sigma > 0$, $\mathcal{N}(\mathbf{x}; \mu, \Sigma)$ is called an “isotropic multivariate normal” or “isotropic Gaussian.”
- If Σ is a positive diagonal matrix, $\mathcal{N}(\mathbf{x}; \mu, \Sigma)$ is called a “diagonal multivariate normal” or “diagonal Gaussian” or “axis-aligned Gaussian.”
- An axis-aligned Gaussian is equivalent to a product of univariate normal marginals.

Factor Analysis

- Factor analysis is a classical statistical model for linear manifolds based on the multivariate normal distribution.
- The model asserts that real-valued data $\mathbf{x}$ with shape $(D, 1)$ are generated in a two stage process that starts by first generating a low-dimensional real-valued latent factor vector $\mathbf{z}$ with shape $(K, 1)$ from a standard multivariate normal distribution.
- The observed $\mathbf{x}$'s are then generated by a diagonal Gaussian with mean given by a linear combination of basis vectors weighted by the latent factor values plus a constant vector: $\mathbf{W}\mathbf{z} + \mu$.
- The matrix $\mathbf{W}$ has size $D \times K$. Each column of $\mathbf{W}$ corresponds to a basis vector.

Factor Analysis: Probabilistic Model

- The probabilistic model/generative process for factor analysis is shown below:

$$p(\mathbf{X} = \mathbf{x}, \mathbf{Z} = \mathbf{z}) = p(\mathbf{X} = \mathbf{x} | \mathbf{Z} = \mathbf{z}) p(\mathbf{Z} = \mathbf{z})$$

$$p(\mathbf{Z} = \mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, I)$$

$$p(\mathbf{X} = \mathbf{x} | \mathbf{Z} = \mathbf{z}) = \mathcal{N}(\mathbf{x}; \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \boldsymbol{\Psi})$$

- $\boldsymbol{\Psi}$ is restricted to be a positive, diagonal matrix. We can learn $\boldsymbol{\mu}$, or simply remove the data set mean and require $\boldsymbol{\mu} = \mathbf{0}$. We will assume the data mean has been removed and thus the optimal value of $\boldsymbol{\mu}$ is $\mathbf{0}$.
- We will first work out the joint distribution for this model, then derive the marginal on $\mathbf{X}$ and conditional on $\mathbf{Z}$ given $\mathbf{X}$.

Factor Analysis: Joint Distribution

To work with the model, we first need to derive the joint distribution of $\mathbf{X}$ and $\mathbf{Z}$.

$$\begin{aligned} p(\mathbf{X} = \mathbf{x}, \mathbf{Z} = \mathbf{z}) &= \mathcal{N}(\mathbf{x}; \mathbf{W}\mathbf{z}, \Psi) \mathcal{N}(\mathbf{z}; 0, I) \\ &\propto \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{W}\mathbf{z})^T \Psi^{-1}(\mathbf{x} - \mathbf{W}\mathbf{z})\right) \exp\left(-\frac{1}{2}\mathbf{z}^T \mathbf{z}\right) \\ &\propto \exp\left(-\frac{1}{2}[\mathbf{x}^T, \mathbf{z}^T] \begin{bmatrix} \Psi^{-1} & -\Psi^{-1}\mathbf{W} \\ -\mathbf{W}^T \Psi^{-1} & \mathbf{W}^T \Psi^{-1} \mathbf{W} + I \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{z} \end{bmatrix}\right) \\ &= \mathcal{N}([\mathbf{x}; \mathbf{z}]; [0; 0], \Sigma) \\ \Sigma^{-1} &= \begin{bmatrix} \Psi^{-1} & -\Psi^{-1}\mathbf{W} \\ -\mathbf{W}^T \Psi^{-1} & \mathbf{W}^T \Psi^{-1} \mathbf{W} + I \end{bmatrix} \end{aligned}$$

Block Matrix Inversion

Block matrix inversion allows us to partition an invertible matrix into four blocks and define the inverse of the matrix in terms of the blocks and their inverses.

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix}$$

Finding the Joint Covariance

$$\begin{aligned}\Sigma &= \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} \Psi^{-1} & -\Psi^{-1}\mathbf{W} \\ -\mathbf{W}^T\Psi^{-1} & \mathbf{W}^T\Psi^{-1}\mathbf{W} + I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix}\end{aligned}$$

$$\begin{aligned}(D - CA^{-1}B)^{-1} &= ((\mathbf{W}^T\Psi^{-1}\mathbf{W} + I) - (-\mathbf{W}^T\Psi^{-1})(\Psi^{-1})^{-1}(-\Psi^{-1}\mathbf{W})) \\ &= ((\mathbf{W}^T\Psi^{-1}\mathbf{W} + I) - (-\mathbf{W}^T\Psi^{-1})\Psi(-\Psi^{-1}\mathbf{W})) \\ &= ((\mathbf{W}^T\Psi^{-1}\mathbf{W} + I) - \mathbf{W}^T\Psi^{-1}\mathbf{W}) \\ &= I\end{aligned}$$

Finding the Joint Covariance

$$\begin{aligned}\Sigma &= \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} \Psi^{-1} & -\Psi^{-1}\mathbf{W} \\ -\mathbf{W}^T\Psi^{-1} & \mathbf{W}^T\Psi^{-1}\mathbf{W} + I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix}\end{aligned}$$

$$\begin{aligned}-A^{-1}B(D - CA^{-1}B)^{-1} &= -(\Psi^{-1})^{-1}(-\Psi^{-1}\mathbf{W})(I) \\ &= \mathbf{W}\end{aligned}$$

Finding the Joint Covariance

$$\begin{aligned}\Sigma &= \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} \Psi^{-1} & -\Psi^{-1}\mathbf{W} \\ -\mathbf{W}^T\Psi^{-1} & \mathbf{W}^T\Psi^{-1}\mathbf{W} + I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix}\end{aligned}$$

$$\begin{aligned}A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} &= (\Psi^{-1})^{-1} + (-\mathbf{W})(-\mathbf{W}^T\Psi^{-1})(\Psi^{-1}) \\ &= \Psi + \mathbf{W}\mathbf{W}^T\Psi^{-1}\Psi \\ &= \mathbf{W}\mathbf{W}^T + \Psi\end{aligned}$$

Finding the Joint Covariance

Putting it all together we have:

$$p(\mathbf{X} = \mathbf{x}, \mathbf{Z} = \mathbf{z}) = \mathcal{N}([\mathbf{x}; \mathbf{z}]; 0, \Sigma)$$

$$\begin{aligned}\Sigma &= \begin{bmatrix} \Psi^{-1} & -\Psi^{-1}\mathbf{W} \\ -\mathbf{W}^T\Psi^{-1} & \mathbf{W}^T\Psi^{-1}\mathbf{W} + I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \mathbf{W}\mathbf{W}^T + \Psi & \mathbf{W} \\ \mathbf{W}^T & I \end{bmatrix}\end{aligned}$$

Factor Analysis: Marginal Distribution

- The marginal distribution of $\mathbf{X}$ can now be found by applying multivariate normal marginalization to the joint distribution. We have:

$$p(\mathbf{Z} = \mathbf{z}, \mathbf{X} = \mathbf{x}) = \mathcal{N}([\mathbf{x}; \mathbf{z}]; 0, \Sigma)$$
$$\Sigma = \begin{bmatrix} \mathbf{W}\mathbf{W}^T + \Psi & \mathbf{W} \\ \mathbf{W}^T & I \end{bmatrix} = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xz} \\ \Sigma_{zx} & \Sigma_{zz} \end{bmatrix}$$

$$\begin{aligned} P(\mathbf{X} = \mathbf{x}) &= \int p(\mathbf{Z} = \mathbf{z}, \mathbf{X} = \mathbf{x}) d\mathbf{z} \\ &= \mathcal{N}(\mathbf{x}; \mu_x, \Sigma_{xx}) \\ &= \mathcal{N}(\mathbf{x}; 0, \mathbf{W}\mathbf{W}^T + \Psi) \end{aligned}$$

Factor Analysis: Learning

- To learn the factor analysis model, we need to minimize the negative log marginal likelihood:

$$\begin{aligned}\text{nlml}(\mathcal{D}, \theta) &= - \sum_{n=1}^N \log \mathcal{N}(\mathbf{x}_n; \mathbf{0}, \mathbf{W}\mathbf{W}^T + \Psi) \\ &= \frac{N}{2} \log(|2\pi(\mathbf{W}\mathbf{W}^T + \Psi)|) + \frac{1}{2} \sum_{n=1}^N \mathbf{x}_n^T (\mathbf{W}\mathbf{W}^T + \Psi)^{-1} \mathbf{x}_n\end{aligned}$$

- Question: To learn the model via direct NLML minimization, what parameter constraints do we need to enforce?

Factor Analysis: Generation/Decoding

- A learned factor analysis model can be used as a *generator*.
- We can choose any vector $\mathbf{z}$, plug it in to the model, and obtain the mean of $p(\mathbf{x}|\mathbf{z})$ as $\mathbf{W}\mathbf{z}$.
- If we want a probabilistic generator, we can sample from $p(\mathbf{x}|\mathbf{z})$.

Factor Analysis: Dimensionality Reduction/Encoding

- A learned factor analysis model can be used as a probabilistic dimensionality reduction model.
- Given a centered value for $\mathbf{x}$, we need to infer the probability distribution on the low-dimensional code $\mathbf{z}$. We have:

Factor Analysis: Conditioning

Since we have the full joint distribution, we can find the conditional distribution on $\mathbf{Z}$ given $\mathbf{x}$ using multivariate normal conditioning:

$$p(\mathbf{Z} = \mathbf{z}, \mathbf{X} = \mathbf{x}) = \mathcal{N}([\mathbf{x}; \mathbf{z}]; \mathbf{0}, \Sigma)$$

$$\Sigma = \begin{bmatrix} \mathbf{W}\mathbf{W}^T + \Psi & \mathbf{W} \\ \mathbf{W}^T & I \end{bmatrix} = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xz} \\ \Sigma_{zx} & \Sigma_{zz} \end{bmatrix}$$

$$p(\mathbf{Z} = \mathbf{z} | \mathbf{X} = \mathbf{x}) = \mathcal{N}(\mathbf{z}; \mu_{z|x}, \Sigma_{z|x})$$

$$\mu_{z|x} = \mu_z + \Sigma_{zx}(\Sigma_{xx})^{-1}(\mathbf{x} - \mu_x) = \mathbf{W}^T(\mathbf{W}\mathbf{W}^T + \Psi)^{-1}\mathbf{x}$$

$$\Sigma_{z|x} = \Sigma_{zz} - \Sigma_{zx}(\Sigma_{xx})^{-1}\Sigma_{xz} = I - \mathbf{W}^T(\mathbf{W}\mathbf{W}^T + \Psi)^{-1}\mathbf{W}$$

Note: $\mu_{z|x}$ is obtained via a *linear* projection from D dimensional space to K dimensional space.

Factor Analysis as an Autoencoder

- The factor analysis model can be used as a particularly complex parameterization of a linear autoencoder trained using maximum likelihood estimation (the equations below assume $\mu = 0$)
- The encoder function is simply the mean of $P(\mathbf{z}|\mathbf{x})$:

$$f(\mathbf{x}) = \mathbb{E}_{P(\mathbf{z}|\mathbf{x})}[\mathbf{z}] = \mathbf{V}\mathbf{x}$$
$$\mathbf{V} = \mathbf{W}^T(\mathbf{W}\mathbf{W}^T + \Psi)^{-1}$$

- The decoder function is the mean of $P(\mathbf{x}|\mathbf{z})$:

$$g(\mathbf{z}) = \mathbb{E}_{P(\mathbf{x}|\mathbf{z})}[\mathbf{x}] = \mathbf{W}\mathbf{z}$$

Autoencoders vs Factor Analysis

- Autoencoders and classical factor analysis can be applied to many of the same tasks including denoising and representation learning.
- Deep autoencoders can learn to represent data from complex manifolds using highly non-linear representations; however, they are not probabilistic models.
- Classical factor analysis is a probabilistic model, but the mean relationship between the latent variables and the visible variables is linear.
- What if we could merge the strengths of these two model classes and build deep, non-linear probabilistic autoencoders?

Non-Linear Factor Analysis

- This model generalizes the basic factor analysis model where the relationship between $\mathbf{z}$ and the mean of $\mathbf{X}$ is linear, to the case where the mean of $\mathbf{X}$ is generated from $\mathbf{z}$ using a deterministic non-linear transformation $f_{\mathbf{w}}(\mathbf{z})$ specified by a neural network.
- As with the basic factor analysis model $p(\mathbf{Z} = \mathbf{z}) = \mathcal{N}(\mathbf{z}; 0, I)$.
- The conditional probability of $\mathbf{X}$ is given by:

$$p(\mathbf{X} = \mathbf{x} | \mathbf{z}, \theta) = \mathcal{N}(\mathbf{x}; f_{\mathbf{w}}(\mathbf{z}), \Psi)$$

- $f_{\mathbf{w}}(\mathbf{z})$ represents a decoder/generator network. The model parameters are $\mathbf{w}$ and Ψ . Ψ is again a diagonal covariance matrix.

Non-Linear Factor Analysis: Joint Density

- The joint density of $\mathbf{X}$ and $\mathbf{Z}$ is thus given by:

$$p(\mathbf{X} = \mathbf{x}, \mathbf{Z} = \mathbf{z} | \theta) = \mathcal{N}(\mathbf{x}; f_{\mathbf{w}}(\mathbf{z}), \Psi) \mathcal{N}(\mathbf{z}; 0, I)$$

- Note that the joint density is computable and the computational complexity depends mostly on the complexity of the generator network $f_{\mathbf{w}}(\mathbf{z})$.
- Also note that given a trained model, we can efficiently generate new data from the model by sampling $\mathbf{z} \sim \mathcal{N}(0, I)$ and then $\mathbf{x} \sim \mathcal{N}(f_{\mathbf{w}}(\mathbf{z}), \Psi)$.
- However, the joint *distribution* of $\mathbf{X}$ and $\mathbf{Z}$ is not longer Gaussian.

Non-Linear Factor Analysis: Learning

- The marginal distribution of $\mathbf{X}$ is given by:

$$\begin{aligned} p(\mathbf{X} = \mathbf{x}|\theta) &= \int p(\mathbf{X} = \mathbf{x}, \mathbf{Z} = \mathbf{z}|\theta) d\mathbf{z} \\ &= \int \mathcal{N}(\mathbf{x}; f_{\mathbf{w}}(\mathbf{z}), \Psi) \mathcal{N}(\mathbf{z}; \mathbf{0}, I) d\mathbf{z} \end{aligned}$$

- To learn the non-linear factor analysis model using maximum likelihood, we need to minimize the negative log marginal likelihood:

$$nlml(\mathcal{D}, \theta) = - \sum_{n=1}^N \log \left(\int p(\mathbf{X} = \mathbf{x}_n, \mathbf{Z} = \mathbf{z}|\theta) d\mathbf{z} \right)$$

- However, the integral in the NLML has no closed-form solution...